LAB ASSIGNMENT : 06

Problem Statement: Write a C function to compute the factorial of a number without recursion.

Objective: To demonstrate two different approaches for calculating the factorial of a number : iterative & recursive, showcasing the versatility of problem-solving methods in programming.

Theory:

1) Explain the difference between the iterative & recursive approaches to calculating the factorial of a number.

A.

Iterative Method	Recursive Method.
a) Uses loops like for or while.	a) A function calls itself to solve smaller subproblems.
b) Repeats statements until condition becomes false.	b) Must have a base case to stop recursion.
c) Does not use function calls	c) Uses stack memory for each function call.
d) Faster and more efficient for large values.	d) May cause stack overflow for large inputs.

2) What is the base case in the recursive function for computing factorials, and why is it necessary?

A. The base case for factorial is:

```
if (n==0 || n==1)
    return 1;
```

a) It stops the recursive function from calling itself infinitely.

b) Without a base case, the program will cause infinite

recursion & crash.
c) It provides the final value from which recursion builds the answer.

3) How does the program handle invalid input, such as negative numbers, and why is this important?

A. If input is negative: a) The program should display an error message. b) It should not calculate factorial.

Ex: if (n<0)
printf("Factorial not defined for negative no's:");

a) It prevents logical errors.
b) It avoids infinite recursion.
c) Improves program reliability.

Platform: Linux

Conclusion: Thus, we have successfully executed factorial of a number with and without recursion.

FAQs:

1) What is the purpose of calculating factorial in programming?

A. Factorial is used in:
a) Permutations & combinations (nP_r, nC_r)
b) Probability calculations.
c) Mathematical algorithms.
d) Series expansions (like Taylor series).

2) What is the difference between recursive & non-recursive (iterative) methods?

4.

Recursive Method	Non-Recursive method
a) Uses a function that calls itself	a) Uses loops like for, while, or do-while.
b) Must have a base case to stop execution.	b) Stops when the loop condition becomes false
c) Code is usually shorter and more mathematical	c) Code may be longer but is usually faster
d) May cause stack overflow for large inputs	d) No risk of stack overflow due to repeated calls.